

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

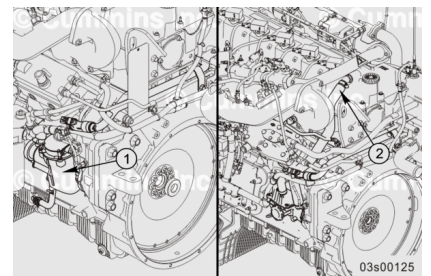
Additional Service Items

- No additional service items required.

General Information

For the electric drive crankcase breather (1), the blowby inlet is integrated in the crankcase breather that is mounted in the cylinder block.

For the air drive crankcase breather (2), the blowby inlet is located in the rocker lever cover.



Preparatory Steps

⚠ WARNING ⚠

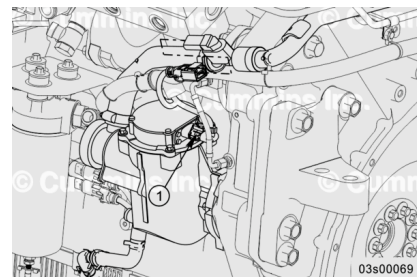
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.

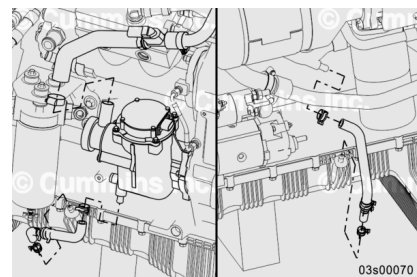
Remove

For the electric drive crankcase breather:

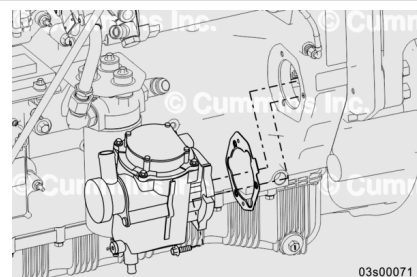
Disconnect the wiring harness connector (1) from the crankcase breather motor.



Disconnect the crankcase breather hoses and oil drain lines.

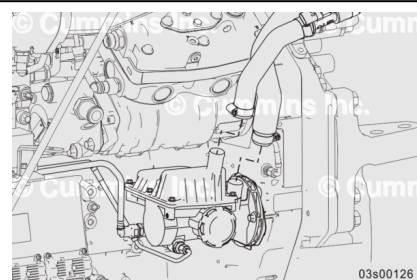


Remove the three capscrews that mount the crankcase breather filter housing. Remove the gasket.

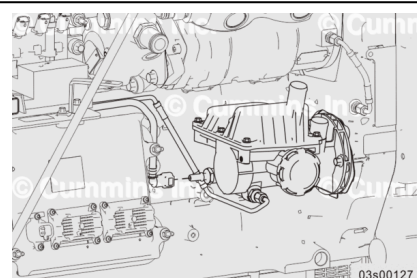


For the air drive crankcase breather:

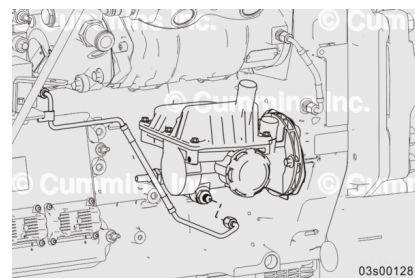
Disconnect the crankcase breather hoses.



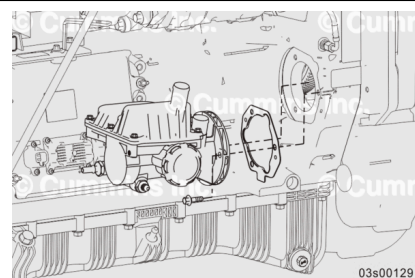
Disconnect the boost air tube.



Remove the oil supply line. Remove the male union if necessary.



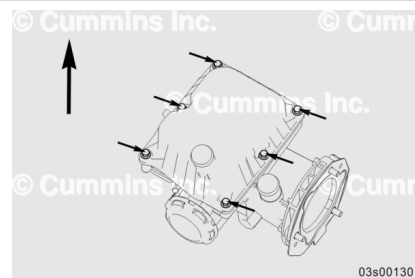
Remove the three capscrews that mount the crankcase breather filter housing. Remove the gasket.



Disassemble

Use the following procedure for the electric drive crankcase breather. Refer to Procedure 003-039 in Section 3.
(</qs3/pubsys2/xml/en/procedures/1016/1016-003-039.html>)

For the air drive crankcase breather, remove the crankcase cover plate capscrews.

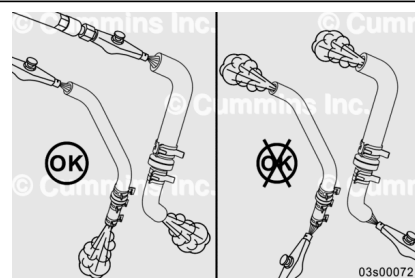


Clean and Inspect for Reuse

Inspect the crankcase ventilation oil drain tube check valves for correct operation.

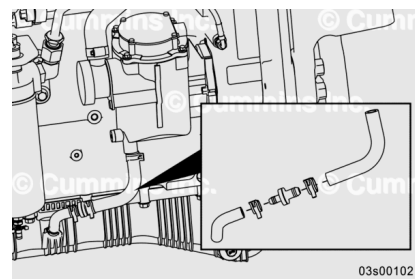
The check valve shall:

- Allow oil to drain from the crankcase ventilation filter housing to the oil pan.
- Prevent crankcase gases from traveling up to the crankcase ventilation filter housing.



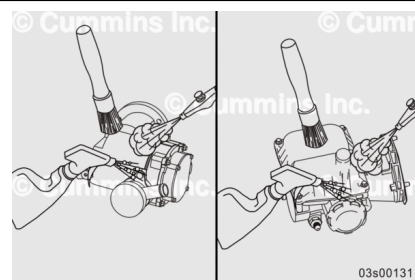
A small amount of air can be blown through the line (less than 34 kPa [5 psi]) to test the operation of the check valve. If the check valve does **not** hold pressure, replace the check valve.

The check valve flows from the thin side to the thick side. The thick side of the check valve is close to the oil pan and the thin side is close to the crankcase breather.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ CAUTION ⚠

Do not submerge the crankcase filter in fluid. This will result in damage to the crankcase ventilation filter.

Clean the exterior surface of the crankcase ventilation filter housing with hot, soapy water.

Dry with compressed air.

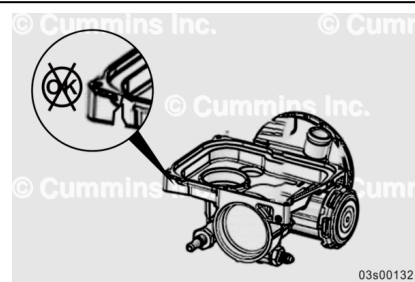
For the air drive crankcase breather, inspect the inside of the crankcase breather for cracks, chippings, or oil.

Replace the entire assembly if any damage is found.

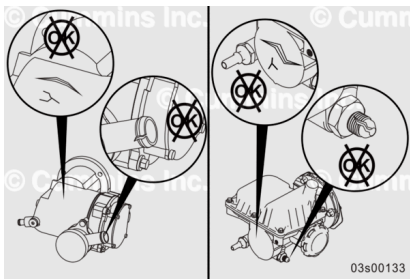
A small amount of oil is acceptable.

Clean the inside of the crankcase breather with clean lubricating oil if any chippings are found.

Dry with compressed air.



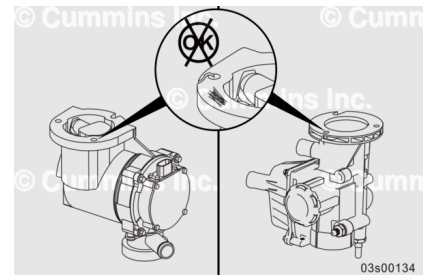
Inspect the crankcase ventilation filter housing for cracks. Replace the entire assembly if any damage is found.



Inspect the sealing surface for damage.

Look for cracks, cuts, swelling, and any other noticeable damage.

Replace if damage is found.

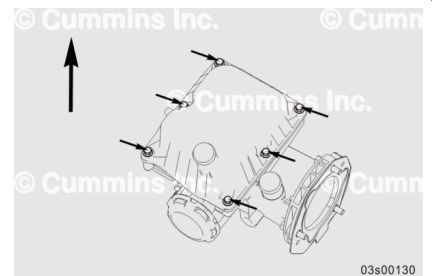


Assemble

Use the following procedure for the electric drive crankcase breather. Refer to Procedure 003-039 in Section 3.
(/qs3/pubsys2/xml/en/procedures/1016/1016-003-039.html)

For the air drive crankcase breather, install the crankcase cover plate capscrews.

Torque Value: 5 n•m [44 in-lb]

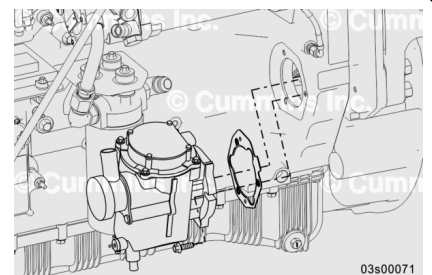


Install

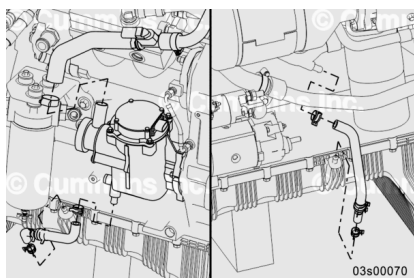
For the electric drive crankcase breather:

Install the gasket and three capscrews that mount the crankcase breather filter housing.

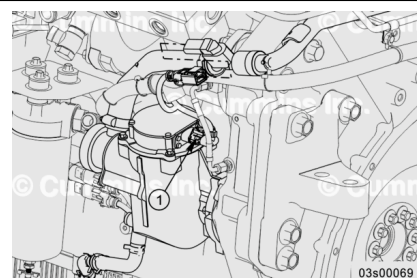
Torque Value: 23 n•m [204 in-lb]



Connect the crankcase breather hoses and oil drain lines.



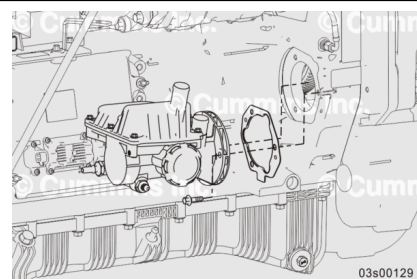
Connect the wiring harness connector to the crankcase breather motor.



For the air drive crankcase breather:

Install the gasket and three capscrews that mount the crankcase breather filter housing.

Torque Value: 24 n•m [212 in-lb]

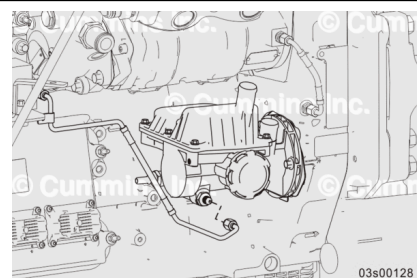


Install the male union, if removed.

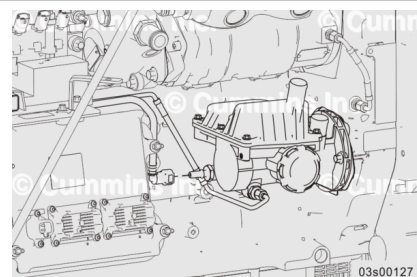
Torque Value: 25 n•m [221 in-lb]

Install the oil supply line to the male union and the crankcase breather.

Torque Value: 20 n•m [177 in-lb]



Connect the boost air tube.

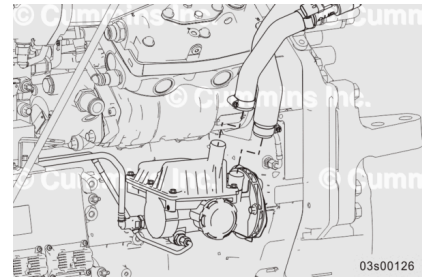


Connect the crankcase breather hoses. Tighten the clamps.

Refer to Procedure 003-024 in Section 3.

(/qs3/pubsys2/xml/en/procedures/1016/1016-003-024-

tr.html)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

If natural gas leaks are present, do not store the equipment inside or in any area that is covered. Severe personal injury can result from asphyxiation or explosion.

- Connect the batteries. See equipment manufacturer service information.